MILITARY SPECIFICATION

TELESCOPE, ELBOW ASSEMBLY: 8211640

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the Telescope, Elbow Assembly, 8211640, part of Aiming Circle M2 and M2A2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-L-10470	-	Lights, Instrument: Fire Control Equipment
MIL-O-13830	-	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-13926	-	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	-	Parts, Equipment, and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling procedures and Tables for Inspection by Attributes

Beneficial comments (recommendations, additions, deletions) and any pertinent date which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development merit Command, Attn, DRDAR-QA, Dover, New Jersey 07801 b using the self-addressed Standardization Document Improvement Proposal (DD Form 1426)appating at the end of this document or by letter.

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

F8211640 Telescope, Elbow Assembly

PACKAGING DATA SHEET

MIL-P-142321P5800554 - Packaginq of Aiminq Circle: M2 W/E

MIL-P-14232/P5911272 - Packaging of Aiming Circle: M2A2 W/E

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The Telescope shall be manufactured in accordance with all drawings applicable to the Telescope specified in the contract or order and drawings pertaining thereto.

3.1.1 First Article Testing. Requirements for the submission of first article samples by the contractor for testing shall be specified in the contract.

3.2 General Specifications. The contractor shall be responsible for adherence to, and compliance with, the following requirements of specifications:

MIL-F-13926.

a. Order of precedence
b. Dimensions and tolerances
c. Inorganic protective surface finishes
d. Commercial quality
e. Part identification and marking
f. Workmanship

3.3 Environmental Service Conditions.

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3.3.1 Storage Temperature. The Telescope shall show no evidence of failure, such as cement separation, glass breakage, or other physical failure, after having been exposed to and thermally stabilized at ambient temperatures of -800 and +160° Fahrenheit (F). Subsequently, the Telescope shall meet the requirements of 3.3.2 through 3.4.7 inclusive.

3.3.2 Vibration. The Telescope shall withstand a vibration of 30 ± 5 cycles per second (CPS) frequency with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes + 15 seconds. Subsequent to vibration, there shall be no visible damage, cement separation, or loosening of assembled parts, and the Telescope shall meet the requirements of 3.4.1 through 3.4.7 inclusive.

3.4 Performance.

3.4.1 Cleanliness. Subsequent to vibration, the Telescope shall meet the cleanliness requirement of MIL-O-13830, and there shall be no more than 2 particles of dirt visible on the surfaces of the reticle. Each particle shall not be in excess of an area equivalent to a size of 1/2 linewidth of the center line of the reticle. There shall be no dirt on other surfaces which interferes with observation or adversely affects image quality.

3.4.2 Sealing. The Telescope shall show no evidence of leakage in excess of 0.2 psig when subjected to an internal pressure of 5.0 ± 0.1 psig for a period of 2 hours using dry nitrogen having a dew point at least as low as -25°F.

3.4.3 Resolution. The resolving power on the Telescope Optical Axis shall be 14 seconds of arc or less.

3.4.4 Eyepiece Focus. The Eyepiece Focus for the Telescope shall be fixed between the limits of -0.75 and -1.0 diopter.

3.4.5 Open Sight. The line of sight of the Open Sight shall be parallel to the line of sight of the Telescope within 10 roils.

3.4.6 Parallax in Telescope. Parallax between reticle and target image shall be nominally zero at 50 yards and shall not exceed 0.30 mil over a range of 30 to 150 yards.

3.4.7 Illumination. The Telescope shall meet the "Illuminated Reticles" requirement of MIL-O-13830. With the Telescope reticle illuminated, the reticle markings shall appear clearly defined when observed in a darkened area.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General Provisions. The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First Article (initial production) Approval, The requirement for first article approval and the responsibility (Government or Contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of (2) Aiming Circles plus (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures, as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.2 Contractor Testing. When the contractor is responsible for conducting First Article Approval Tests, the sample shall be inspected-by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor’s inspection.

4.3 Inspection Provisions.

4.3.1 Submission of Product. Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation? and presentation of lots shall be in accordance with "Submission of product” provisions of MIL-STD-105.

4.3.2 Examination and Tests. Components and subassemblies shall be inspected in accordance with the insection provisions contained in Supplementary Quality Assurance Provisions (SQAP)

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listed in the Technical Data Package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of PSIL-F-13926 shall apply. Examinations and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105, and the sampling plans specified in Tables I and II herein. Examinations and tests for Packaging, Packing, and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects shall constitute the minimum inspection to be performed by the supplier prior to Government acceptance or rejection by lot.

TABLE I. Classification of Defects

Class	Characteristic	Requirement	Test Procedure
Critical:	None		
Major:	AQL 0.65% Defective		
101.	Cleanliness of Telescope	3.4.1	4.6.3
102.	Sealing	3.4.2	4.6.4
103.	Resolution	3.4.3	4.6.5
104.	Eyepiece Focus	3.4.4	4.6.6
105.	Open Sight	3.4.5	4.6.7
106.	Parallax in Telescope	3.4.6	4.6.8
107.	Illumination	3.4.7	4.6.9
Minor:	None		

4.3.3 Acceptance and Rejection. Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Functional tests. The requirements and tests in Table 1 shall be inspected on a 100% basis. All examinations and tests shall be conducted at the standard ambient temperature (600 to 90°F).

4.4 Special Sampling.

4.4.1 General and Environmental. Three Telescopes, Elbow shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table II.

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TABLE II

No.	Characteristic	Requirement	Test Procedure
201.	Fabrication	3.1	Applicable drawings, MIL-L-1047O, Visual
202.	General Specification	3.2	MIL-F-13926, Visual
203.	Storage Temperature	3.3.1	4.6.1
204.	vibration	3.3.3	4.6.2

4.4.2 Failure of Sample. Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3) .

4.5 Inspection Equipment. Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government Furnished Inspection Equipment. Where the contract Provides for Government furnished test equipment, supply and maintnance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor Furnished Inspection Equipment.

4.5.2.1 Government Design. All inspection equipment, specified by drawing number in specifications or SQAP'S forming a part of the contract, shall be supplied by the contractor in accordance with technical data included in the Technical Data Package list.

4.5.2.2 Contractor Design. The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the Component Inspection Procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the

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prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test Methods and Procedures.

4.6.1 Storage Temperature. The Telescope shall be exposed to the temperatures specified in 3.3.1, utilizing testing equipment conforming with the "Test Facilities" requirements of MIL-STD-810. In the performance of this test, the temperatures shall be changed gradually to avoid thermal shock. After temperature stabilization at -800 and +160°F, the Telescope shall be exposed to each specified temperature for a minimum of 5 hours. Subsequent to being exposed to the above conditions, and upon return to the standard ambient temperature of +60° and +90°F, the Telescope shall be subjected to a visual and tactile examination to assure that there is no evidence of mechanical and optical failure, cement separation? or deterioration. The unit shall then be tested in accordance with 4.6.2 through 4.6.9 to determine compliance with 3.3.2.

4.6.2 Vibration. The Telescope shall be subjected to the vibration test utilizing standard vibrating equipment capable of meeting the conditions of 3.3.2. The accuracy of the applicable test equipment and test equipment measurement shall be in accordance with "Test Facilities" requirements of MIL-STD-810. The Telescope shall be mounted in its normal operating position on the vibrating table by means of an adapter. The Telescope shall be vibrated in a vertical plane with the motion and frequency applied for the time period specified. At the conclusion of this vibration, the Telescope shall be subjected to a visual and tactile examination for damage such as bent or broken parts, glass chips, fractures, foreign matter, and cement separation; and to a functional examination of moving parts for undue irregularities of movement, friction, or looseness. Any one of the aforementioned defects shall be cause to consider a Telescope defective unless separate defects-per-hundred type sampling plans are established or approved by the Government as part of the supplier's inspection system. Telescopes that have passed the above examinations shall be subjected to the tests of 4.6.3 through 4.6.9 inclusive, in compliance with 3.3.3.

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4.6.3 Cleanliness. The optical surfaces of the Telescope shall be inspected for cleanliness in accordance with MIL-0-13830. The Telescope reticle shall not contain particles of dirt in excess of the amount specified in 3.4.1. Area inspection of dirt particles shall be accomplished by means of a Scratch and Dig Standard conforming to Drawing C7641866 for a comparison of the area of the dirt particles with the dig size of the standard. The Telescope shall conform to its respective cleanliness requirement. This test shall be performed subsequent to vibration.

4.6.4 Sealing. Sealing tests for the Telescope shall be performed utilizing special testing equipment conforming to the Two Station-Pressure Tester, Drawing F8565556, wherein a special hose adapter to pressurize the Telescope will be required. The procedure for the internal pressure test described in the set-up and operating instructions on Sheet 1 of Drawing F8565556 shall apply. The Telescope shall be pressurized with a dry nitrogen gas to the required pressure and maintained for the time specified. Leakage shall not be evident within the time period specified for sealing.

4.6.5 Resolution. To perform this test, a wall target containing the appropriate resolution target shall be employed. Construction of the resolution target and the method of inspection shall outlined in MIL-O-13830. The telescope must resolve the test pattern within the limits outlined in 3.4.3. During this test, the telescope may be supported in a “V” block. The use of packing or shims may be used to position the telescope. When making this test, use a dioptometer having a minimum of 4 power.

4.6.6 Eyepiece Focus. The eyepiece focus test shall be Performed with the aid of a dioptometer conforming to the Diopto-meter, Drawing F7680631. The dioptometer shall be adjusted by setting the slide scale index to zero, and the eyepiece of the dioptometer shall be adjusted for the best focus on the diopto-meter reticle. Place the dioptometer at the eyepiece of the Telescope on the special testing equipment. Adjust the diopto-meter, by movement of the dioptometer tube, until the Telescope reticle appears in sharp focus. The diopter scale on the tube of the dioptometer shall indicate between the -0.75 (3/4) and -1.0 diopter to conform with 3.4.4.

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4.6.7 Open Sight. The open sight shall be checked with the same set-up employed for the test in 4.6.5. With the Telescope focused on the target's line of sight mark, the open sight shall read within the limits of the circle on the target described in the set-up instruction, Drawing 7687113, to determine compliance with the requirements specified in 3.4.5.

4.6.5 Parallax in Telescope. With the Elbow Telescope mounted on the special testing equipment and leveled, parallax shall be checked by sighting through the ocular of the Telescope and observing a point of coincidence on a target, adjusted to the distance specified in 3.4.6, with respect to a corresponding point of the Telescope's reticle in close-proximity to the Telescope's optical axis. With the target image adjusted to the 50-yard range, parallax between the reticle and target image shall be parallax free as viewed through the eyepiece assembly of the Telescope. The target shall contain lines subtending 0.3 mils right and left, UP and down from the original coincidence setting. Parallax will be recognized as any apparent displacement of the target's image in relation to the reticle image when the observer's head is moved from side to side or up and down. Any apparent movement of the target image shall not exceed the target tolerance lines at the distance specified in 3.4.6.

4.6.9 Illumination. The illumination test for the requirements of 3.4.7 is a visual check, and shall be conducted in accordance with MIL-O-13830. Illumination of the reticle shall be accomplished with the lighting instrument energized with a 3-volt source.

5. PACKAGING.

5.1 Packaging, Packing, and Marking. Packaging, packing, and marking shall be in accordance with MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P5800554 for M2 Aiming Circle and Packaging Data Sheet MIL-P-14232\P5911272 for M2A2 Aiming Circle. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended Use. Telescope, Elbow Assembly: 8211640 is intended for use as components of Aiming Circle M2 and M2A2.

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6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Selection of an applicable level of protection, packaging, and packing.
c. Applicable stock number.
d. Applicable packaging data sheet number (see 5.1).
e. Requirements for submittal of first article samples.

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